The centroid G divides the median AD internally in the ratio 2:1 and therefore by section formula, the centroid

$$G(x,y) \text{ is } \left(\frac{\frac{2(x_2+x_3)}{2}+1(x_1)}{2+1}, \frac{\frac{2(y_2+y_3)}{2}+1(y_1)}{2+1} \right) = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$$

The centroid G of the triangle with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is

$$G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$$

Activity - 2

1. Draw $\varnothing ABC$ with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ on the graph sheet.

2. Draw medians and locate the centroid of $\varnothing ABC$

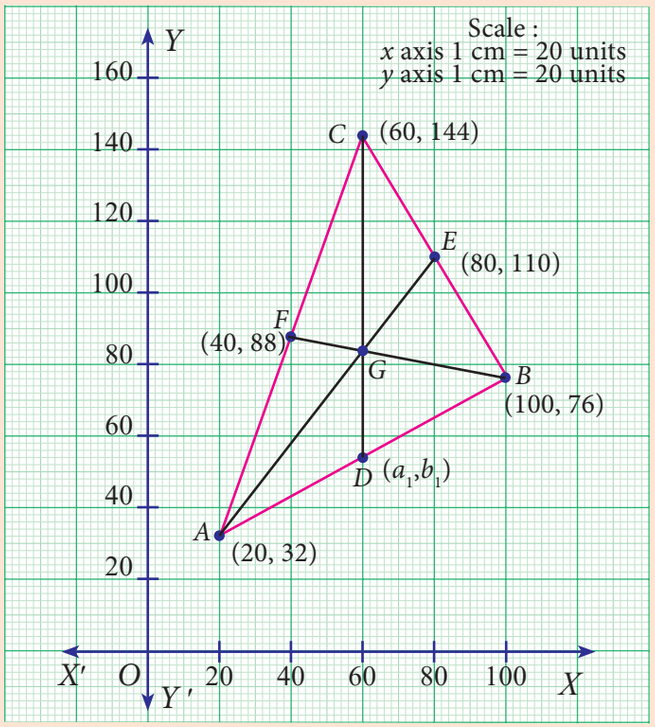


Fig. 5.38

Observation

(i) The coordinates of the vertices of $\varnothing ABC$ where $A(x_1, y_1) =$ _____, $B(x_2, y_2) =$ _____ and $C(x_3, y_3) =$ _____

(ii) The coordinates of the centroid G= _____

(iii) Use the formula to locate the centroid, whose coordinates are = _____.

(iv) Mid-point of AB is _____.

(v) Find the point which divides the line segment joining (x_3, y_3) and the mid-point of AB internally in the ratio 2:1 is _____.

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